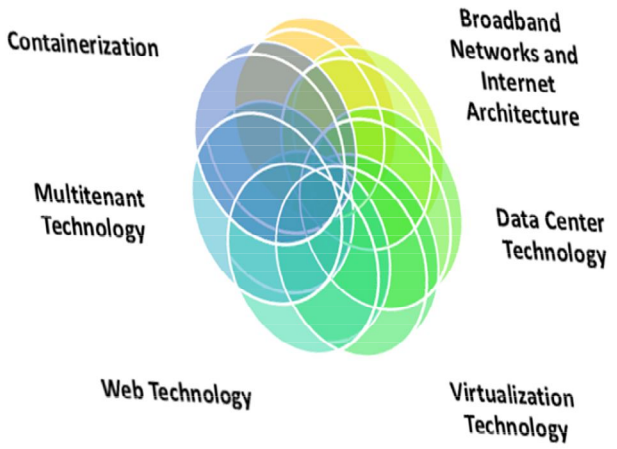
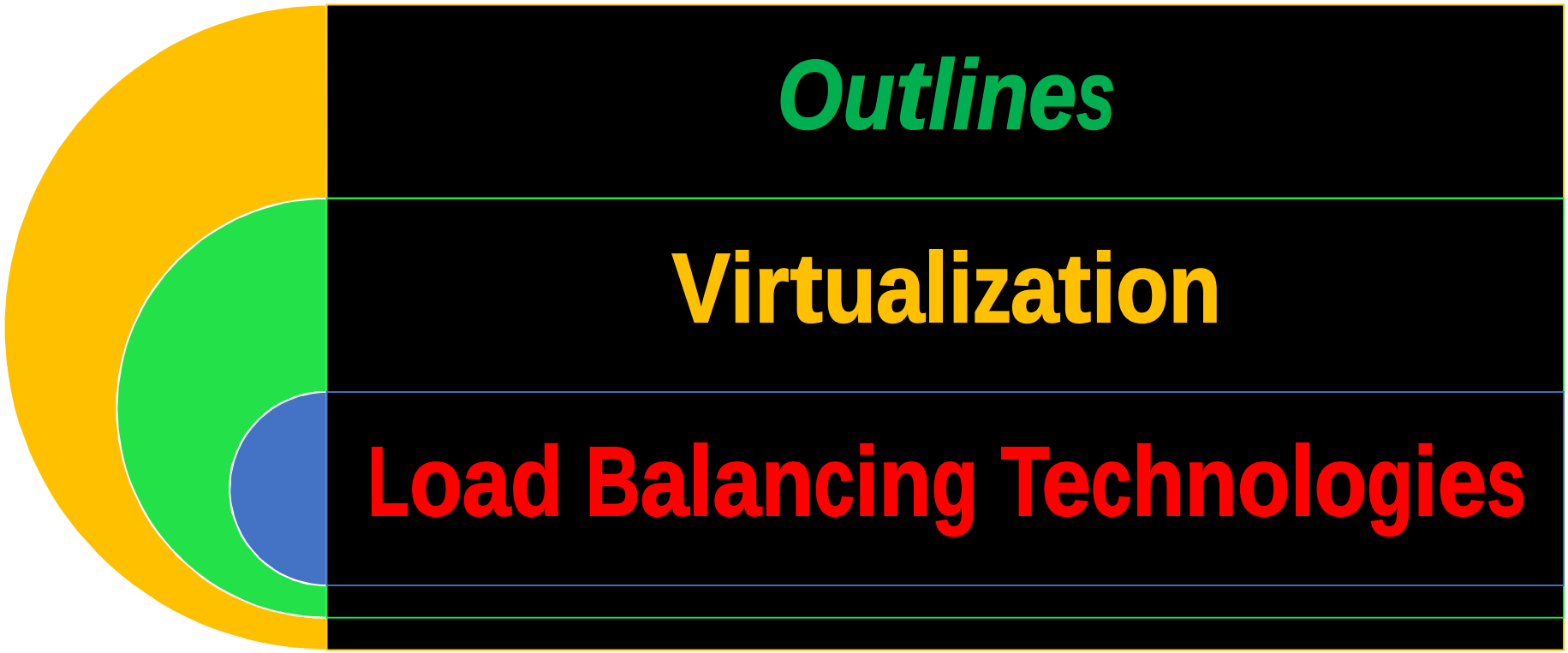
Chapter 4

Cloud-Enabling Technologies



By Tagel W.(M.Sc.)

tagel.12wogayehu@gmail.com


Outlines
Virtualization
Load Balancing Technologies


Containerization
Broadband Networks and Internet Architecture
Multitenant Technology
Cloud Computing Enabling Technologies
Data Center Technology
Web Technology
Load Balancing Technology
Virtualization Technology

Broadband Networks and Internet Architecture

Broadband Networks and Internet Architecture

- All clouds must be connected to a network; this **inevitable** requirement forms an inherent dependency on internetworking.
- Internetworks, or the Internet, allow for the remote provisioning of IT resources and are directly supportive of ubiquitous network access.
- Cloud consumers have the option of accessing the cloud using only private and dedicated network links in LANs, although most clouds are Internet enabled.
- The potential of cloud platforms therefore generally grows in parallel with advancements in Internet connectivity and service quality

Broadband Networks and Internet Architecture

- The Internet's topology has become a dynamic and complex aggregate of ISPs that are highly interconnected
- **Internet Service Providers (ISPs)** can freely deploy, operate, and manage their networks in addition to selecting partner ISPs for interconnection.
- **Advancement in packet switching technology and router interconnectivity are** used to construct the internetworking architecture this ISPs.

Internet Service Providers (ISPs)

- No centralized entity comprehensively governs the Internet, although bodies like the Internet Corporation for Assigned Names and Numbers (ICANN) supervise and coordinate Internet communications.
- Governmental and regulatory laws dictate the service provisioning conditions for organizations and ISPs both within and outside of national borders.
- The Internet's topology has become a dynamic and complex aggregate of ISPs that are highly interconnected via its core protocols.
- Smaller branches extend from these major nodes of interconnection, branching outwards through smaller networks until eventually reaching every Internet-enabled electronic device.

Connectionless Packet Switching

- End-to-end (sender-receiver pair) data flows are divided into packets of a limited size that are received and processed through network switches and routers, then queued and forwarded from one intermediary node to the next.
- Each packet carries the necessary location information, such as the Internet Protocol (IP) or Media Access Control (MAC) address, to be processed and routed at every source, intermediary, and destination node.

Router-Based Interconnectivity

- A router is a device that is connected to multiple networks through which it forwards packets.
- Even when successive packets are part of the same data flow, routers process and forward each packet individually while maintaining the network topology information that locates the next node on the communication path between the source and destination nodes.
- Routers manage network traffic and gauge the most efficient hop for packet delivery, since they are privy to both the packet source and packet destination.

Router-Based Interconnectivity

- The communication path that connects a cloud consumer with its cloud provider may involve multiple ISP networks.
- The Internet's mesh structure connects Internet hosts (endpoint systems) using multiple alternative network routes that are determined at runtime.
- Communication can therefore be sustained even during simultaneous network failures, although using multiple network paths can cause routing fluctuations and latency.

Data Center Technology

Data Center Technology

- A data center is a facility made up of networked computers, storage systems, and computing infrastructure that businesses and other organizations use to organize, process, store large amounts of data.
- A business typically relies heavily on applications, services, and data within a data center, making it a focal point and critical asset for everyday operations.

Data Center Technology

- A data center facility enables an organization to assemble its resources and infrastructure for data processing, storage, and communication, including:
  - o systems for storing, sharing, accessing, and processing data across the organization;
  - o physical infrastructure to support data processing and data communication; And
  - o Utilities such as cooling, electricity, network access, and uninterruptible power supplies (UPS).
- Gathering all these resources in one data center enables the organization to:
  - o protect proprietary systems and data;
  - o Centralizing IT and data processing employees, contractors, and vendors;
  - o Enforcing information security controls on proprietary systems and data; And
  - o Realize economies of scale by integrating sensitive systems in one place.

Data Center Technology

- **A modern data center concentrates an organization's data systems in a well-protected physical infrastructure, which includes:**
  - ***Server;***
  - ***storage subsystems;***
  - ***networking switches, routers, and firewalls;***
  - ***cabling; And***
  - ***Physical racks for organizing and interconnecting IT equipment.***
- Data Center Technologies
  - ***Data Center Consolidation***
  - ***Data Center Colocation***

Data Center Consolidation

- There is no need for a single data center.
- Modern businesses can use two or more data center installations in multiple locations for greater flexibility and better application performance, reducing latency by locating workloads closer to users.
- Consolidating data center facilities offers the ability to enhance operations, improve redundancy and flexibility, deliver higher availability and better use of IT assets.

Data Center Colocation

- Datacenter operators may also pay a fee to rent server space in a colocation facility. A colocation is an attractive option for organizations that want to avoid the large capital expenditure associated with building and maintaining their data centers.

Virtualization

Virtualization

- Virtualization is the creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources.
- In other words, Virtualization is a technique, which allows to share a single physical instance of a resource or an application among multiple customers and organizations.
- It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.
- Creation of a virtual machine over existing operating system and hardware is known as Hardware Virtualization.
- A Virtual machine provides an environment that is logically separated from the underlying hardware.
- The machine on which the virtual machine is going to create is known as Host Machine and that virtual machine is referred as a Guest Machine.

Types Virtualization

Hardware Virtualization

- When the virtual machine software or virtual machine manager (VMM) is directly installed on the hardware system is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.
- After virtualization of hardware system we can install different operating system on it and run different applications on those OS.
- Hardware virtualization is mainly done for the server platforms, because controlling virtual machines is much easier than controlling a physical server.

Types Virtualization

Operating system Virtualization

- When the virtual machine software or virtual machine manager (VMM) is installed on the Host operating system instead of directly on the hardware system is known as operating system virtualization.
- Operating System Virtualization is mainly used for testing the applications on different platforms of OS.

Types Virtualization

Server Virtualization

- When the virtual machine software or virtual machine manager (VMM) is directly installed on the Server system is known as server virtualization.
- Server virtualization is done because a single physical server can be divided into multiple servers on the demand basis and for balancing the load.

Types Virtualization

Storage Virtualization

- Storage virtualization is the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.
- Storage virtualization is also implemented by using software applications and it is used for mainly done for back-up and recovery purposes.

Load Balancing Technology

Load Balancing

- Load balancing is the sharing of the loads between different nodes for effectively giving services to the cloud user without hampering the underlying infrastructure.
- It is the distribution of loads over the different nodes which provides good resource utilization when nodes are overloaded with job.
- Load balancing is a technique that ensures that no VM will be idle while other VMs are being utilized
- Load balancing is a technique that is used to balance the load between the nodes or VM of the cloud.
- Balancing means that it makes the less loaded node take additional loads and makes the highly loaded node to be free from taking or accepting additional loads.

Load Balancing

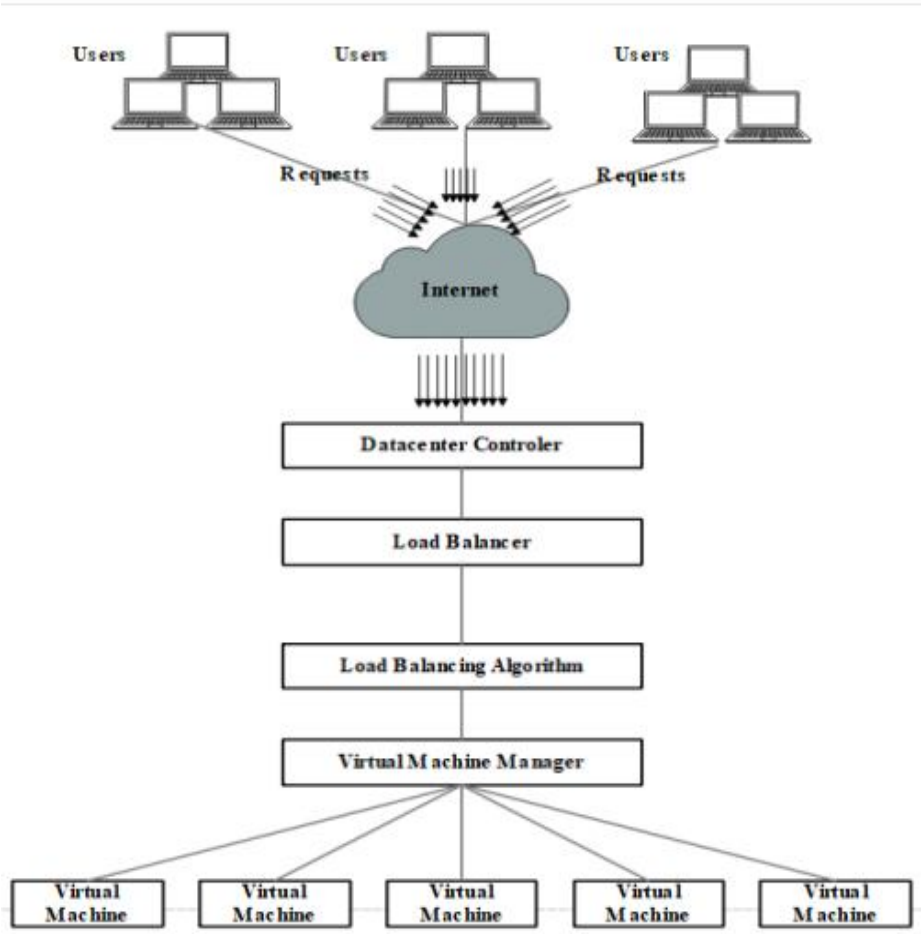
- Load balancing has several important goals which are mentioned in different studies. Following are listed some of them.
  - To maintain system stability
  - To improve system performance
  - To improve availability in cloud system
  - To make cloud systems to be available for future modification and enhancement
  - To have a backup plan in case of system failure
  - To reduce and minimize job response time
  - To enhance and improve resource utilization
- However The main goal of load balancing is to have higher resource utilization with minimal response time and minimum datacenter (DC) processing time.

Load Balancing

- Two major kinds of load balancers can be used for balancing loads in cloud environments.
- The first one is a hardware load balancer and the other is a software load balancer.
- Hardware-based load balancers are load balancers that are placed in front of the servers and manage the loads' route based on the system performance. Hardware-based load balancers are very expensive and use application-specific hardware-dependent components.
- Software load balancers can run in each machine in a cluster, if one node in a cluster goes down the other nodes in that cluster are alerted by communication and take the additional load in addition to theirs.

Load Balancing

- The load balancing is done by using a load balancing algorithm (LBA).
- LBA is a method that is implemented by the load balancers to balance the load in the cloud VMs.
- In cloud computing, LBAs are used to manage and distribute the loads across the VMs evenly.
- From architecture when we see this LBA they are found in between the cloud load balancer and the VM manager.
- Initially, the different requests that are sent by the CSC or the cloudlets are received by the DC controller. The DC controller controls the DC activities.
- The load balancer is used to implement the different policies that are used to manage and balance loads



Load Balancing Algorithms

- In cloud computing, LBAs are algorithms that are used to balance the loads that are forwarded to the nodes.
- Most likely CLBAs are considered as the scheduling algorithms that are used to manage and schedule resource usage in the operating system. However, it's obvious and clear that these algorithms work for balancing a higher amount of load which is far greater than the personal computers operating system.
- There are several kinds of LBA in which many studies have categorized them into two major broad categories saying that
    - Static and
    - Dynamic LBAs

Static Cloud Load Balancing Algorithms

- Static CLBAs are algorithms that are used to balance loads of mainly small-scale cloud systems.
- Static CLBAs do not consider the initial state of the system. I.e. loads can be assigned to a working or busy VM equally with the other VMs.
- The distribution of the load between the idle VM, less loaded VM and the overloaded VM is equal if they are in a similar cloud system.
- The static types of CLBAs are mainly used for response time minimization in small-scale clouds.
- In the best case where all the first VMs are always free, the static algorithms perform excellent with minimal response time unlikely if it has to be in the worst case when all the VMs are overloaded the response time will take a considerable time until the VMs become free and give the response to the request in the queue.
- LBAs like round-robin are good examples of this kind of algorithms.

Dynamic Cloud Load Balancing Algorithms

- The dynamic CLBAs depend on the prior knowledge of the VMs state. But unlike the static CLBAs Dynamic CLBAs check the state of each node before assigning a load to the respective node.
- By using a dynamic LBA the state of the node is first checked either it is an overloaded or busy state in that case if the node is busy the request will be transferred to the other node.
- Here, the algorithm will not stop until it gates the free node. Once after it has got it will pass the request to that node and goes to handle other request or job.
- The main disadvantage of this dynamic CLBA is that it has added overhead for searching the free node from the other nodes in the connection.
- This has a big effect at the time of worst-case when if all the nodes are handling a specific job for a long burst time the algorithm may check for the free node repeatedly.
- Unlike the static CLBAs, these dynamic CLBAs are used to minimize response time and have higher throughput in both large and small cloud systems.

Load Balancing Algorithms

Some Basic CLBAs

- Round Robin Algorithm
- Randomized Algorithm
- Min-Min Algorithm
- Max-Min Algorithm
- Throttled Algorithm
- Active VM Monitoring Algorithm (ESCE)
- Threshold Algorithm
- Honey Bee Algorithm
- Ant colony Algorithm

Cloud Analyst

- Deploying the LBA in real cloud applications or systems and actual performance testing in the real cloud environment requires a wide range of cloud infrastructures.
- Since setting up this cloud infrastructure is very expensive and demands a large budget the experiments and tests in cloud environment (CLBAs) are done by using a cloud simulation tool that has similar simulation capability for the implementation and experimentation of the CLBAs like the real cloud environment.
- Cloud analyst is a simulation tool that is extended and built on CloudSim and it is used to simulate cloud applications with different deployment configurations.
- *Cloud analyst is developed in Australia, University Melbourne by Bhatiya Wickremasinghe and Rajkumar Buyya at CLOUDS laboratory of the computer science and software engineering department.*

Cloud Analyst



Cloud Analyst

- Cloud analyst has several promising features that it's very useful for testing CLBAs with its package.
- Cloud analyst offers a user to add personalized load balancing policy or LBA to its package to test and evaluate its response time, DC processing time and other metrics.
- In Cloud analyst after simulation of a certain application, the response time will be presented in different ways. I.e. the minimum, maximum and average response time will be stated. This is because of the difference in the geographical location of the users accessing the cloud application.

Cloud Analyst

- The other benefit of this Cloud analyst simulator is, it is user-friendly.
- Components in Cloud analyst are presented in the GUI (graphical user interface) mode.
- Users can add VMs, add DCs, and configure the user bases and other components by using the GUI of the simulator.
- In addition to this by using Cloud analyst, it is possible to extract the simulation and experimental outputs inform of PDF formats.
- By using the Cloud analyst tool it is possible to do repeated experiments by saving the experiment's input parameters and results in the form of XML.

Further Reading Assignments on Cloud Enabling Technologies

- Multitenancy technology
- The web technologies and the web services including the REST
- The Advancement on computer (Servers) Processing capability
- Mobile Computing (Advancement on smartphones)
- Containerization
- 3G, 4G and the 5G networks

END

End of chapter 4

Qn. if Any???

